

Ziyun (Claude) Wang

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ziyunclaudewang.github.io | Google Scholar: scholar.google.com/citations?user=mctHCc0AAAAJ

EDUCATION

GRASP Lab, University of Pennsylvania 2025

Ph.D., Computer Science

Advisor: Dr. Kostas Daniilidis

Beyond Frames: Learning to Perceive with Event-Based Vision

GRASP Lab, University of Pennsylvania 2019

M.S.E., Robotics

Motion Equivariant Networks for Event Cameras with the Temporal Normalization Transform

Rice University 2017

B.S., Computer Science

EXPERIENCE

Assistant Professor August 2025 – present

Electrical and Computer Engineering

Data Science and AI Institute

Laboratory for Computational Sensing and Robotics

Johns Hopkins University

Research Intern May 2024 – August 2024

Vision Product Group (VPG), Apple

- Developed 3D computer vision algorithms for enhancing the viewing experience under large motion on the Vision Pro.

Research Intern August 2019 – June 2020

Samsung AI Center New York

- Developed novel single-view 3D reconstruction algorithms for robotics.
- Developed cost-to-go learning algorithms for efficient motion planning.

PUBLICATIONS

1. Ruijun Zhang, Hang Su, Kostas Daniilidis, **Ziyun Wang**. “Match-Any-Events: Zero-Shot Motion-Robust Feature Matching Across Wide Baselines for Event Cameras.” *European Conference on Computer Vision (ECCV)* **2026**
2. **Ziyun Wang**, Ruijun Zhang, Zi-Yan Liu, Yufu Wang, Kostas Daniilidis. “Continuous Time Human Motion Field from Events”. *International Conference on Computer Vision (ICCV)* (**Acceptance rate: 24%**) **2025**
3. Royina Karegoudra Jayanth*, Yinshuang Xu*, **Ziyun Wang**, Evangelos Chatzipantazis, Kostas Daniilidis, Daniel Gehrig. “EqNIO: Subequivariant Neural Inertial Odometry” *International Conference on Learning Representations (ICLR)* **2025**
4. **Ziyun Wang**, Jinyuan Guo, Kostas Daniilidis. “Un-EVIMO: Unsupervised Event-based Independent Motion Segmentation.” *European Conference on Computer Vision (ECCV)* (**Acceptance rate: 24.9%**) **2024**
5. Friedhelm Hamann, **Ziyun Wang**, Ioannis Asmanis, Kenneth Chaney, Guillermo Gallego, Kostas Daniilidis. “Motion-prior Contrast Maximization for Dense Continuous Time Motion Estimation” *European Conference on Computer Vision (ECCV)* (**Acceptance rate: 24.9%**) **2024**
6. Yufu Wang, **Ziyun Wang**, Lingjie Liu, Kostas Daniilidis. “TRAM: Global Trajectory and Motion of 3D Humans from in-the-wild Videos.” *European Conference on Computer Vision (ECCV)* (**Acceptance rate: 24.9%**) **2024**
7. Yunzhou Song, Jiahui Lei, **Ziyun Wang**, Lingjie Liu, Kostas Daniilidis. “Track Everything Everywhere Fast and Robustly” *European Conference on Computer Vision (ECCV)* (**Acceptance rate: 24.9%**) **2024**
8. **Ziyun Wang***, Kenneth Chaney*, Kostas Daniilidis. “EvAC3D: From Event-Based Apparent Contours to 3D Models via Continuous Visual Hulls.” *European Conference on Computer Vision (ECCV)* (**Oral Presentation, 2.7% of submissions**) **2022**
9. **Ziyun Wang***, Fernando Cladera*, Anthony Bisulco, Daewon Lee, Camillo J Taylor, Kostas Daniilidis, M Ani Hsieh, Daniel D. Lee, Volkan Isler. “EV-Catcher: High-Speed Object Catching Using Low-Latency Event-Based Neural Networks.” *IEEE Robotics and Automation Letters (RA-L)* **2022**
10. Alex Zhu, **Ziyun Wang**, Kaung Khant, Kostas Daniilidis. “Eventgan: Leveraging large scale image datasets for event cameras.” *IEEE International Conference on Computational Photography (ICCP)* **2021**

11. Jinwook Huh, Galen Xing, **Ziyun Wang**, Volkan Isler, Daniel D. Lee. "Learning to generate cost-to-go functions for efficient motion planning." *Experimental Robotics: The 17th International Symposium* **2021**
12. **Ziyun Wang**, Eric Mitchell, Volkan Isler, Daniel D. Lee. "Geodesic-HOF: 3D Reconstruction Without Cutting Corners." *AAAI Conference on Artificial Intelligence (Acceptance rate: 21.4%)* **2021**
13. **Ziyun Wang**, Volkan Isler, Daniel D. Lee. "Surface HOF: Surface Reconstruction from a Single Image Using Higher Order Function Networks." *IEEE International Conference on Image Processing (ICIP)* **2020**

UNDER REVIEW

1. Linli Shi, Ruijun Zhang, **Ziyun Wang**. "EVIS: A Physics-Grounded Event Camera Plugin for NVIDIA Isaac Sim." *arXiv preprint* **2026**
2. Ziyue Zheng*, Linli Shi*, Bingkun He, Wen Jiang, **Ziyun Wang**. "TRACE: Ergodic Trajectory Optimization for Active Scene Reconstruction." *arXiv preprint* **2026**

OPEN-SOURCE SOFTWARE

1. **EVIS**: Physics-Grounded Event Camera Plugin for NVIDIA Isaac Sim, v1.0. Developed with Linli Shi and Ruijun Zhang. **2026**

SELECTED REFEREED WORKSHOP PAPERS

1. **Ziyun Wang**, Friedhelm Hamann, Kenneth Chaney, Wen Jiang, Guillermo Gallego, Kostas Daniilidis. "Event-based Continuous Color Video Decompression from Single Frames." *CVPR Event-based Vision Workshop*. **2025**
2. Kenneth Chaney*, Fernando Cladera*, **Ziyun Wang**, Anthony Bisulco, M Ani Hsieh, Christopher Korpela, Vijay Kumar, Camillo J Taylor, Kostas Daniilidis. "M3ED: Multi-Robot, Multi-Sensor, Multi-Environment Event Dataset." *Event Vision Workshop, CVPR* **2023**
3. Alex Zhu, Wenxin Liu, **Ziyun Wang**, Vijay Kumar, Kostas Daniilidis. "Robustness Meets Deep Learning: An End-to-End Hybrid Pipeline for Unsupervised Learning of Egomotion." *Workshop on Deep Learning for Semantic Visual Navigation, CVPR* **2019**

PATENTS

1. **Ziyun Wang**, Eric Anthony Mitchell, Ibrahim Volkan Isler, and Daniel D. Lee. "Method and apparatus for three-dimensional (3D) object and surface reconstruction." U.S. Patent 11,380,061, issued July 5, 2022.

2. **Ziyun Wang**, Fernando Cladera Ojeda, Anthony Robert Bisulco, Dae Won Lee, Camillo J. Taylor, Konstantinos Daniilidis, Ani Hsieh, and Ibrahim Volkan Isler. “Systems and methods for real-time state estimation of fast-moving objects.” U.S. Patent Application 17/978,873, filed May 4, 2023.
3. Jinwook Huh, Galen Kailun Xing, **Ziyun Wang**, Ibrahim Volkan Isler, and Daniel D. Lee. “Trajectory generation of a robot using a neural network.” U.S. Patent 11,642,787, issued May 9, 2023.

SELECTED PRESENTATIONS AND PANELS

Event-Driven Robotic Vision, August 2026, Inaugural Speaker, APL–JHU Collaboration Seminar Series, Johns Hopkins University Applied Physics Laboratory, MD

Tutorial: Event-Based Computer Vision, June 2026, Telluride Neuromorphic AI Workshop, Telluride, CO

Event-Driven Robotic Vision, January 2026, LCSR Seminar, Johns Hopkins University, MD

Beyond Frames: Learning to Perceive with Event-Based Vision, November 2025, Robotics Seminar, Worcester Polytechnic Institute, MA

Industry vs. Academia at P³ Workshop, October 2025, Panel, University of Delaware, DE

Event-Driven Computer Vision, November 2025, Machine Perception guest lecture, Johns Hopkins University, MD

Beyond Frames: Learning to Perceive with Event-Based Vision, June 2025, Student Presentation, University of Pennsylvania, PA

Towards Bio-Inspired Efficient AI: Rethinking Perception and Action with Neuromorphic Vision, February 2025, Computer Science Seminar, Stony Brook University, NY

Towards Bio-Inspired Efficient AI: Rethinking Perception and Action with Neuromorphic Vision, March 2025, ECE Seminar, Johns Hopkins University, MD

Continuous-time Human Motion Fields from Event Cameras, February 2025, NYC Computer Vision Day, New York City

Continuous Color Video Decompression from Single Frames, February 2025, Research Talk, Apple, CA

High-Speed Object Catching Using Low-Latency Event-based Neural Networks, Nov 2021, Center for Brain-Inspired Computing (C-BRIC), Purdue University, IN

High-Speed Object Catching Using Low-Latency Event-based Neural Networks, Aug 2021, Neuromorphic Seminar Invited Talk, University of Pennsylvania, PA

Self-supervised Geometric Learning for Flow, Depth and Odometry, June 2019, Research Talk, Samsung AI Center, NY

TEACHING AND SERVICE

Area Chair 2027
Winter Conference on Applications of Computer Vision (WACV)

Co-organizer 2026
Workshop on Perception and Decision Making for Athletic Humanoid Robotics (IROS 2026)
Telluride Neuromorphic AI Workshop (June 2026)
Workshop on Challenges and Opportunities of Neuromorphic Field Robotics and Automation (ICRA 2026, Vienna)

Instructor 2026
EN.520.681 Perception for Embodied AI (Spring 2026)
EN.520.665 Machine Perception (Fall 2026)

Member 2025
ECE Faculty Hiring Committee

Outstanding Reviewers
CVPR 2025, ECCV 2024

Reviewing 2021–present
CVPR, ECCV, ICCV, AAAI, TPAMI, IROS, ICRA, RA-L, TRO, NeurIPS, Nature Communications

Teaching Assistant
Machine Perception (CIS 580), Dr. Kostas Daniilidis, University of Pennsylvania
Advanced Robotics (MEAM 620), Dr. Camillo J. Taylor and Dr. Ani Hsieh, University of Pennsylvania